



ELKATECH LAUNCHPAD

Product Requirements Document

ElkaTech Launchpad — Service Platform

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1. Product Overview

ElkaTech Launchpad is a single monorepo that delivers two connected products for ElkaTech, an industrial printing and signage machinery business:

- A **public marketing and product-catalog website** that presents the ElkaTech brand, machine categories, and product detail, with a polished dark/light themed experience and stable, layout-safe motion.
- An **authenticated service portal** where customers raise and track machine service requests, and where staff (engineers, support, owners, admins) triage, assign, and resolve that work.

The two products share one React/Vite frontend, one Fastify gateway, four internal services (auth, catalog, service-desk, notification), shared TypeScript/Zod contracts, PostgreSQL, Firebase Authentication, and Cloudflare R2 for attachment storage. Role-based permissions are defined once in the shared contracts and enforced on the backend gateway as well as reflected in the UI.

2. Business Goals

- Present ElkaTech's machinery portfolio professionally and convert public interest into service relationships.
- Give existing customers a reliable, self-serve channel to report machine problems against the specific machine they own.
- Let operational staff run service delivery — triage, assignment, status, and customer communication — in one place.
- Keep ownership and access controls strict: customers see only their own data; destructive actions are limited to administrators.
- Remain low-cost and easy to operate (serverless deployment on Vercel, hosted Postgres on Neon, object storage on Cloudflare R2).

3. Target Users and Personas

PERSONA	DESCRIPTION	PRIMARY NEEDS
Customer	Owns one or more ElkaTech machines	Report a problem on the right machine, attach evidence, track progress
Engineer	Field/technical staff	See queue, claim/assigned work, update status, message the customer
Support	Service-operations staff	Monitor queue, look up customer activity, create requests for customers, assign engineers

PERSONA	DESCRIPTION	PRIMARY NEEDS
Owner	Business operator	Operational oversight plus user approval, suspension, and role assignment (except admin)
Admin	System administrator	Full control, including permanent deletion and the system health/danger panel

All five roles exist in the current branch and are documented as implemented.

4. Core Problems Being Solved

- Customers had no structured way to report machine issues tied to the exact unit, location, and serial they own.
- Staff lacked a shared queue with assignment, status workflow, and an auditable history of who did what and when.
- The business needed clear separation between **catalog products** (what is sold) and **customer machines** (what a specific customer actually owns).
- Access needed real backend enforcement, not just hidden buttons, with a clear rule that only admins may permanently delete data.

5. Product Scope

In scope:

- Public website and product-category discovery.
- Customer authentication, onboarding profile, and machine-scoped requests.
- Request lifecycle: creation, attachments, messaging, assignment, status, and history.
- Customer-machine inventory management by admin/owner.
- Operational dashboards (Customer Activity, Support) for support/owner/admin.
- User/account management with approval, suspension, role assignment, and admin-only hard delete.
- Transactional email notifications.

Out of scope is listed in Section 17.

6. Public Website Requirements

- Responsive landing page with hero, about, work, brands, infrastructure, why-us, contact, and footer sections.
- Product-category pages for solvent printers, UV printers, laser cutting machines, lamination machines, desktop UV printers, inkjet printers, and flatbed UV printers.

- Per-product image carousels, specification tables, brochure links, and a clear entry point into the service portal.
- Persisted light/dark/system theme with theme-aware navigation.
- Entrance reveals must be layout-stable (no refresh-time shift); honour `prefers-reduced-motion`.

7. Service Portal Requirements

- Cookie-based authenticated sessions with CSRF protection on mutations.
- Sign-up/sign-in via Firebase (email/password and Google), bridged to a server-side session; a legacy email/password path remains for existing accounts.
- New public customers begin in `pending_approval` and cannot create requests until an admin or owner approves them.
- Customers must complete a service profile (company, contact, address) before creating requests.
- Role-aware navigation and routes; each protected route is gated server-side.
- Operational dashboards for support/owner/admin and a system dashboard for admin.

8. Customer Machine Assignment Requirements

- A **catalog product** is the general model offered for sale; a **customer machine** is one physical unit a specific customer owns.
- Only admin/owner can link machines to customers; customers never create or edit machine records.
- Each machine carries a label, optional unit number, site name/location, contact phone, purchase/install dates, an admin-only internal serial number, status (`active` / `inactive`), and a snapshot of the catalog product at assignment time.
- Customers may create requests **only for their own active machines**.
- Admin/owner can create requests on behalf of a customer for that customer's machine; support can also create on behalf using a customer-safe machine view.
- Machines are deactivated (soft `inactive` status), not hard-deleted, so they can be reactivated and their request history is preserved.

9. Service Request Lifecycle

- A request captures the machine, an issue type, description, priority, contact details, and a generated request number.
- Status moves through `new`, `triaged`, `assigned`, `in_progress`, `waiting_for_customer`, `resolved`, and `closed`, enforced by a backend transition map.
- Engineers can self-claim unassigned queue items; support/owner/admin can assign or reassign a specific engineer.

- A request has a conversation (customer-visible messages and internal notes) and an activity history of key events.

10. Attachment / File Upload Requirements

- Customers and staff can attach photo/video evidence to a request.
- Bytes upload **directly from the browser to Cloudflare R2** using a short-lived presigned URL; only metadata (object key, file name, type, size, kind) is stored in the database.
- Uploads are validated for content type and size; when R2 is not configured the presign endpoint returns **501** and the UI degrades gracefully.
- Reads use a configured public base URL when present, otherwise a short-lived presigned read URL.

11. Role-Based Access Requirements

Permissions are enforced on the gateway (returning **403**) and mirrored in the UI. Summary:

CAPABILITY	ADMIN	OWNER	SUPPORT	ENGINEER	CUSTOMER
Permanently delete user data	Yes	No	No	No	No
Approve / reject users	Yes	Yes	No	No	No
Suspend / reactivate users	Yes	Yes	No	No	No
Change roles	Yes (any)	Yes (not admin)	No	No	No
Assign / reassign requests	Yes	Yes	Yes	claim only	No
Create request for a customer	Yes	Yes	Yes	No	own only
View Customer Activity / Support	Yes	Yes	Yes	No	No
Customer machine management	Yes	Yes	view only	No	No
System Admin panel	Yes	No	No	No	No

A non-negotiable rule: **only admins may permanently delete user/customer data**. Owner and support receive a **403** ("Only admins can permanently delete user data.") even on direct API calls.

12. Admin / Staff Workflows

- Invite customer or staff users (owner cannot invite admins).
- Approve, reject, suspend, or reactivate accounts.
- Change a user's role (owner is limited to non-admin roles and may never modify an admin account).
- Link/manage customer machines and create requests on behalf of customers.
- Monitor the queue, assign engineers, and watch system health (admin only).

13. Customer Workflows

- Sign up / sign in, verify email, await approval.
- Complete the required service profile.
- Pick one of their active machines, choose an issue type, describe the problem, attach evidence, and submit.
- Track status, exchange messages, and view the resolution.
- Manage their own account: update display name and profile, reset password.

14. Engineer, Support, and Owner Workflows

- **Engineer:** work the queue, claim or handle assigned requests, update status, and message the customer.
- **Support:** monitor the Support dashboard, look up Customer Activity, create requests for customers, and assign/reassign engineers — without any user-management powers.
- **Owner:** all support operational abilities plus user approval, suspension, role assignment (non-admin), and customer-machine management — but no permanent deletion and no system Admin panel.

15. Notification Expectations

- Auth, machine, and request events are written to per-schema outbox tables.
- The notification service polls those tables and sends transactional emails (account/approval, request created/claimed/assigned, status changes), recording one delivery row per recipient for idempotent retries.
- Locally, Mailpit captures email; in production, SMTP credentials are supplied via environment variables.

16. Non-Functional Requirements

- **Security:** cookie sessions, CSRF on mutations, shared internal-service token, server-side RBAC, presigned (not public-write) storage, admin-only deletion, and protected system/bootstrap accounts.

- **Reliability:** outbox-based notifications with retry idempotency; soft deactivation preserves history; graceful storage fallback.
- **Auditability:** request history records key actions and actors; user state transitions are timestamped.
- **Maintainability:** shared Zod contracts and permission helpers as a single source of truth; small focused services behind one gateway.
- **Deployment:** serverless on Vercel with prebuilt `api/*.mjs` bundles, hosted Postgres (Neon), and Cloudflare R2.

17. Out-of-Scope Items

- Public e-commerce / online machine purchasing and payments.
- Customer self-service machine registration (machines are admin-controlled).
- Inventory, spare-parts, or field-scheduling/dispatch logistics.
- In-app real-time chat or push/mobile notifications beyond email.
- Multi-tenant separation beyond the single ElkaTech organization.

18. Success Metrics

- Share of service issues submitted through the portal versus phone/email.
- Median time from request creation to assignment and to resolution.
- Approval turnaround for new customer accounts.
- Percentage of requests with attached evidence at creation.
- Reduction in mis-routed requests through machine-scoped creation.

19. Risks and Assumptions

- **R2 CORS / configuration:** direct browser uploads require correct bucket CORS; misconfiguration surfaces as failed uploads.
- **Hosted database branching:** preview environments must share the primary database (branch-per-deployment disabled) or migrations will not reach them.
- **Deployment protection:** server-to-server calls require preview SSO protection to be relaxed.
- **Firestore dependency:** sign-in availability depends on correct Admin SDK and web configuration.
- **Assumption:** ElkaTech staff manage approvals and machine assignments promptly so customers are not blocked from creating requests.